

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #23: Hyperlocal Courier Dispatch & Tracking Engine

Use-Case Flow Specification

Use Case	UC-01 – Create Courier Request
Primary Actor	Sender Client

Preconditions

1. The sender client is registered and has access to the courier service.
2. The sender has valid pickup and destination details.
3. The courier service is available to process new requests.

Postconditions

1. A courier request is successfully created.
2. The request is assigned to an active delivery rider within the specified 3 km radius.
3. The sender receives confirmation of the courier request.

Main Success Scenario

1. The Sender Client selects the option to create a new courier request.
2. The system displays the courier request form.
3. The sender enters the pickup location and destination details.
4. The system validates the entered pickup and destination information.
5. The system creates the courier request.
6. The system identifies the nearest active delivery rider within a 3 km radius.
7. The system assigns the courier request to the selected rider.
8. The system sends a dispatch notification to the Delivery Rider.
9. The system displays a confirmation of the assigned courier request to the Sender Client.
10. The use case ends successfully.

Alternate Flow – No Eligible Rider Available

1. At Step 6, the system cannot find an active delivery rider within the 3 km radius.
2. The system informs the Sender Client that no eligible rider is currently available.
3. The courier request is placed in a pending state.
4. The Sender Client may retry the request or cancel it.
5. The use case ends when the request is retried successfully or cancelled.